Adarsha Aryal

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LinkedIn | GitHub | Website

Summary

Master of Data Science graduate with a background in engineering and business. I have hands-on experience building real-time data pipelines, scalable data systems, and analytics solutions. I work with Python, R, SQL, PySpark, Kafka, and AWS to design and deploy end-to-end data solutions. I am passionate about transforming raw data into reliable infrastructure and meaningful insights that create real impact.

Experience

Project Engineer at China Communications Services
Kathmandu, Nepal | October 2019 - December 2023

- Designed and implemented 4G network and power infrastructure projects supporting Nepal Telecom across multiple regions.
- Optimised 4G network infrastructure planning by analysing 100+ technical survey reports, reducing design errors by 15% through rigorous data validation and quality auditing.
- Enhanced budget accuracy by performing complex cost estimation and financial evaluations, effectively minimising cost overruns across multiple regional projects.

Project Engineer at Airflow Industries Pvt. Ltd.
Kathmandu, Nepal | September 2017 - October 2019

- Designed electrical power systems and HVAC layouts for pharmaceutical and commercial clients, ensuring compliance with industry standards.
- Streamlined project delivery timelines by monitoring installation and commissioning data, reducing delays through proactive resource allocation.
- Increased project profit margins by up to 10% by conducting financial analysis and cost-optimisation modelling for project proposals.

Side Projects

Hive Joy (Australian Honey Marketplace)
- Built an end-to-end platform with Supabase authentication, PostgreSQL database, seller workflow, and role-based access control and deployed in Vercel.
- Built a synthetic marketplace dataset and performed EDA using Python (Pandas, NumPy) to analyze GMV, seller performance, and customer behavior.

Weather Watch (Global Weather Analytics Pipeline)
- Built an hourly Databricks pipeline processing 1.1M rows across 100 cities using PySpark and Delta Lake (Bronze → Silver → Gold).
- Developed anomaly detection, K-means clustering and Prophet forecasting models tracked via MLflow, served through a live Streamlit app.

Data Visualisation Project (US Mass Shootings Dashboard)
- Used open datasets to perform exploratory analysis.
- Built dynamic, interactive dashboards using R-shiny.

Industry Project (CoolRoute)
- Worked with team of five to build an Android application.
- Recommends the most comfortable routes for cyclists/pedestrians, rather than fastest route.

Education

Master of Data Science
Feb 2024 – Dec 2025
Monash University, Melbourne, Australia
Machine Learning, Big Data, Cloud Computing

Master of Business Administration Aug 2019 – Mar 2023
Pokhara University, Pokhara, Nepal
Accounting, Finance, Marketing

Work Rights In Australia

Full-Time up to March 2028
(Graduate Visa- 485)

Languages

Nepali, English (88/90 in PTE)

Volunteering Experience

Worked as a Peer Mentor for Juniors while doing my final year in undergrad.

Worked as the President of the Electrical Club in undergrad for 2 years.

References can be made available upon request.